

1 Overview

The DWM3001CDK is a development board that facilitates evaluation of the DWM3001C module. Figure 1 highlights the main components and interfaces.

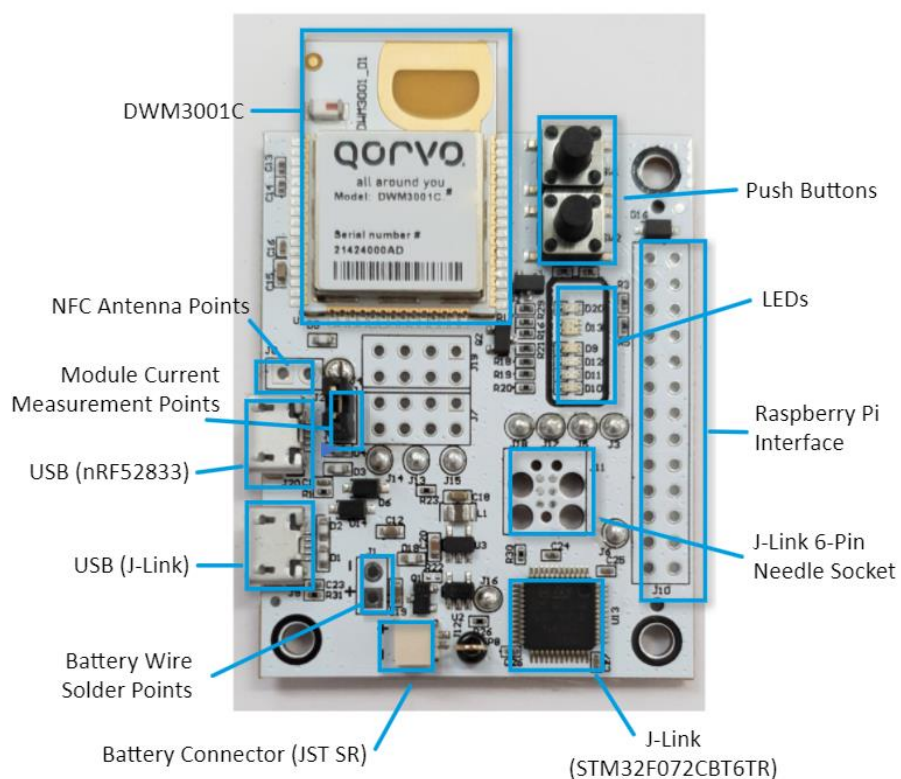


Figure 1. Overview of DWM3001CDK

2 Power Options

The DWM3001CDK can be powered via USB, battery, or Raspberry Pi. The connector numbers are listed in Table 1. The board does not include a battery charger.

Power Option	Connector on PCB
USB – J-Link	J9
USB – nRF52833	J20
Battery – JST SR Connector	J12
Battery – Loose Wires	J1
Raspberry Pi Interface	J10

Table 1. DWM3001CDK Power Options

3 Solder Bridge Jumpers

Table 2 lists the solder bridge jumpers on the development board. These can be desoldered to facilitate making current consumption measurements on the module, with peripherals such as LEDs and STM MCU isolated. The STM MCU runs the Segger J-link OB firmware.

Solder Bridge	Purpose	Default State
J3	Supply rail connection for LEDs. Desolder to remove power to LEDs, e.g. for minimizing current consumption.	Soldered
J4	Series bridge on DWM3001C module supply rail. Desolder to measure module current consumption at J2.	Soldered
J5	Ground connection for dual-LED (D13) circuit. Desolder to remove power to LED, e.g. for minimizing current consumption.	Soldered
J6	Series bridge on Segger RESET signal. Desolder to isolate IO leakage from current consumption measurement.	Soldered
J14	Series bridge on Segger TXD signal. Desolder to isolate IO leakage from current consumption measurement.	Soldered
J15	Series bridge on Segger RXD signal. Desolder to isolate IO leakage from current consumption measurement.	Soldered
J16	Series bridge on Segger MCU supply rail. Desolder to isolate MCU from current consumption measurement.	Soldered
J17	Series bridge on Segger SWDIO signal. Desolder to isolate IO leakage from current consumption measurement.	Soldered
J18	Series bridge on Segger SWDCLK signal. Desolder to isolate IO leakage from current consumption measurement.	Soldered

Table 2. Solder bridges and their purposes

4 Raspberry Pi Interface

The DWM3001CDK has an interface to allow connection to a Raspberry Pi board. A connector such as the stacking header shown in Figure 2 can be attached at J10 for this purpose.



Figure 2: Stacking Header for Raspberry Pi (e.g. Adafruit 1112)

Figure 3 shows how the DWM3001CDK board can be mounted on a Raspberry Pi board using the stacking header. The J10 pins are simply aligned with pins 1 to 26 of the Raspberry Pi header.

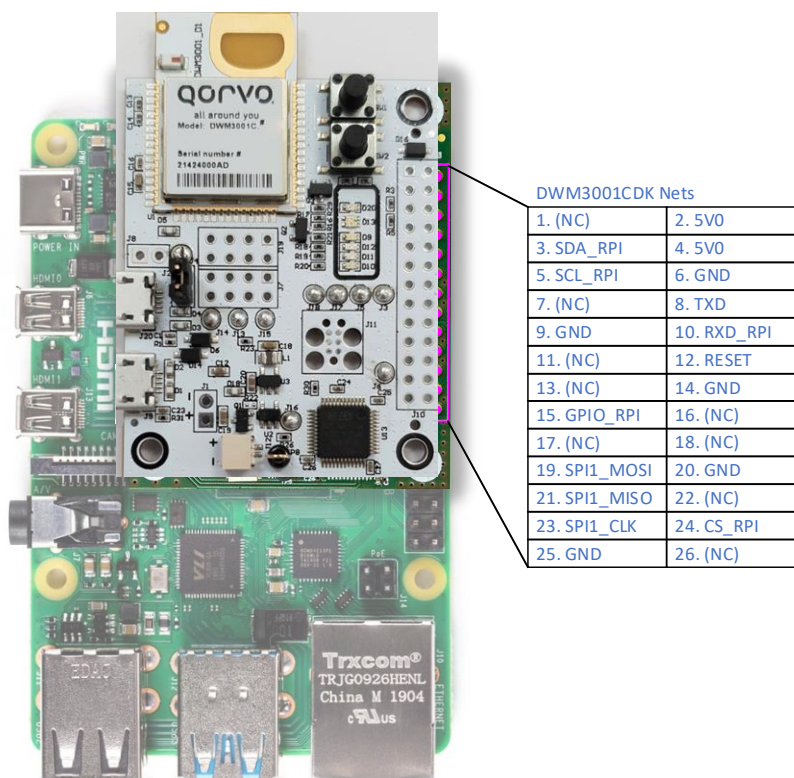
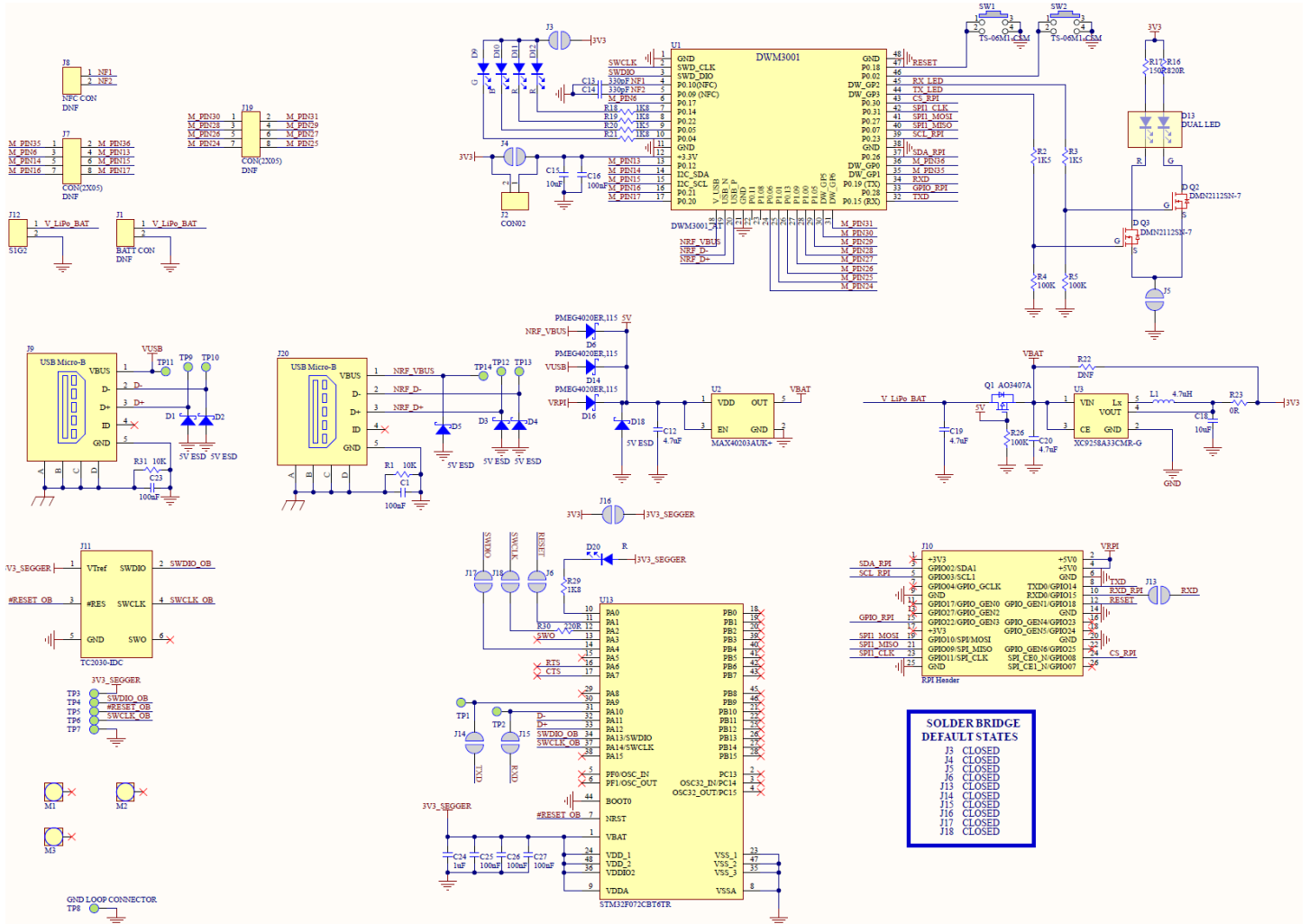


Figure 3. DWM3001C Mounted on Raspberry Pi

8 Schematic



9 Antenna Delay Calibration

To measure range accurately, precise calculation of timestamps is required. To do this the antenna delay must be known. The DWM3001C allows this delay to be calibrated and provides the facility to compensate for delays introduced by PCB, external components, antenna and internal DWM3001C delays.

To calibrate the antenna delay, range is measured at a known distance using two DWM3001C systems. Antenna delay is adjusted until the known distance and reported range agree. The antenna delay can be stored in OTP memory.*

***Note: For Engineering Samples DWM3001C.E1.0SR, DWM3001C.E1.0SQ, DWM3001C.E1.0SB and DWM3001CDKE1.0: do not use the Channel 5 Antenna Delay in OTP – use default value 16390.**